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Introduction and Overview



THE GRID (Electrical Power System)

- ◆ Earth by night
 - <https://svs.gsfc.nasa.gov/5276/>
 - High light density – highly industrialized areas
 - Low light density – Rural area

- ◆ Energy can be grouped into
 - Mechanical energy
 - Light
 - Heat
 - Electrical energy
 - Chemical energy

Balanced Production and Consumption

- ◆ Different types of loads
 - Residential loads
 - Commercial and industrial loads
 - Electrical railways

- ◆ Loads are demanding from the grid
 - Active and reactive power
 - Constant frequency
 - Constant voltage

- ◆ Grid demands power generating stations

Electrical Power Production

- ◆ Primary sources of the power generators
 - Hydro electric power
 - Fossil fuels (Thermal power)
 - Natural gas
 - Crude oil
 - Coal
 - Uranium (Nuclear thermal power)
 - Renewable energy
 - Sunlight
 - Wind
 - Tidal water flow
 - Geothermal energy
 - Biomass

Electrical Power Transmission

- ♦ Energy Transmission
 - Connect loads and energy production
- ♦ Large power plants
 - Provide bulk power to the high-voltage (HV) transmission network
- ♦ Distribute power between consumers (loads)
 - Medium voltage (MV) distribution network

Electrical Power Systems – History

- ◆ Power system grow over time
- ◆ In Vorarlberg driven by industrialization
 - 1820 and 1830
 - Production facilities has been established
 - Coal has to be imported to drive the machinery
 - 1884 and 1890
 - Textual manufacturer installed an operated hydro electric power plants (F.M. Hämmerle, Ganahl, Getzner)
- ◆ 1891
 - First three-phase transmission line between Neckar and Frankfurt am Main, Germany

Electrical Power Transmission – History

- ◆ 1908
 - Hydro power plant Andelsbuch (15 MW)
 - Biggest power plant in Austro-Hungarian Empire
- ◆ 1916
 - E-Werke Jenny & Schindler Rieden becomes Vorarlberger Kraftwerke GmbH Andelsbuch
- ◆ 1924
 - Founding of Vorarlberger Illwerke GmbH
- ◆ 1925
 - Arlberg railway become fully electrified (Put into operation 1884)

Electrical Power Transmission – History

◆ 1929

- First 220 kV transmission line between two countries from Vorarlberg, Austria to Braunweiler, Germany
- Distance 600 km
- Milestone in the European Grid history

◆ 1930

- Pump storage hydro plant Vermuntwerk put into operation

Electrical Power Systems – History until Today

- ♦ In the beginning individual grids
 - Generation and consumption together
 - E.g. within an production plant
- ♦ Today we see a similar trend
 - Distributed energy resources (DER)
 - <https://energiegemeinschaften.gv.at/>
- ♦ Generation capacity = Maximum possible load
- ♦ No use of capacity if load is below maximum

Electrical Power Systems – Load Sharing

- ◆ Isolated systems
 - Generation capacity = Maximum possible load
 - No use of capacity if load is below maximum

- ◆ Interconnected systems
 - Load / generation capacity sharing
 - More efficient

Electrical Power Systems – Load Sharing

♦ Example

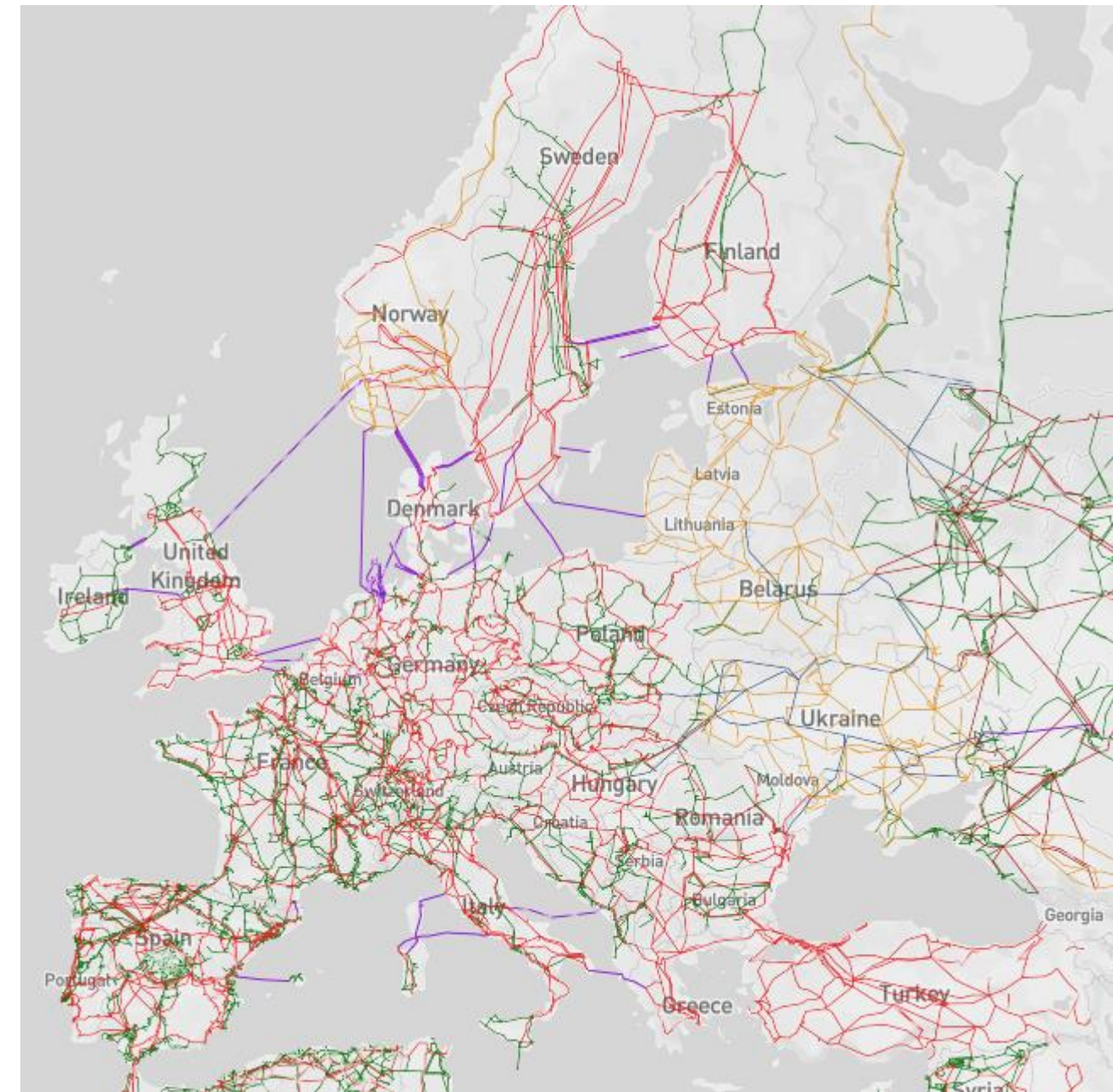
- Household 1 and 2
 - washer, dryer regular household – 2 kW
- Scenario 1
 - Household 1 and 2 have their own 2 kW generator
- Scenario 2
 - Interconnect household 1 and household 2
 - Manage to distribute peak demands
 - Only one generator needed

Electrical Power System - Vorarlberg

- ◆ Transmission system operator (TSO) in Vorarlberg
 - Vorarlberger Energienetze GmbH
- ◆ Connections to neighboring electrical power systems
 - 220 kV and 380 kV line to Germany (TransnetBW). First line 1929, second line 1958
 - 380 kV between Bürs and Austrian Power Grid (APG)
 - 220 kV between Meiningen and Swissgrid

Electrical Power System - Europe

- ♦ European Network of Transmission System Operators for Electricity (ENTSO-E)
 - Organization of European TSO's
 - ENTSO-E itself is connected with its neighbors
 - Interconnection map [ENTSO-E Transmission System Map](#)



Electrical Power Transmission – Disadvantages

- ◆ Must be sufficient interconnection transport capacity
- ◆ Power losses because of energy exchanges
- ◆ In large systems parallel transport can put higher constraints on the system. E.g. Energy flow via Poland instead within Germany North to South
- ◆ Organizational matters are required in order to operate

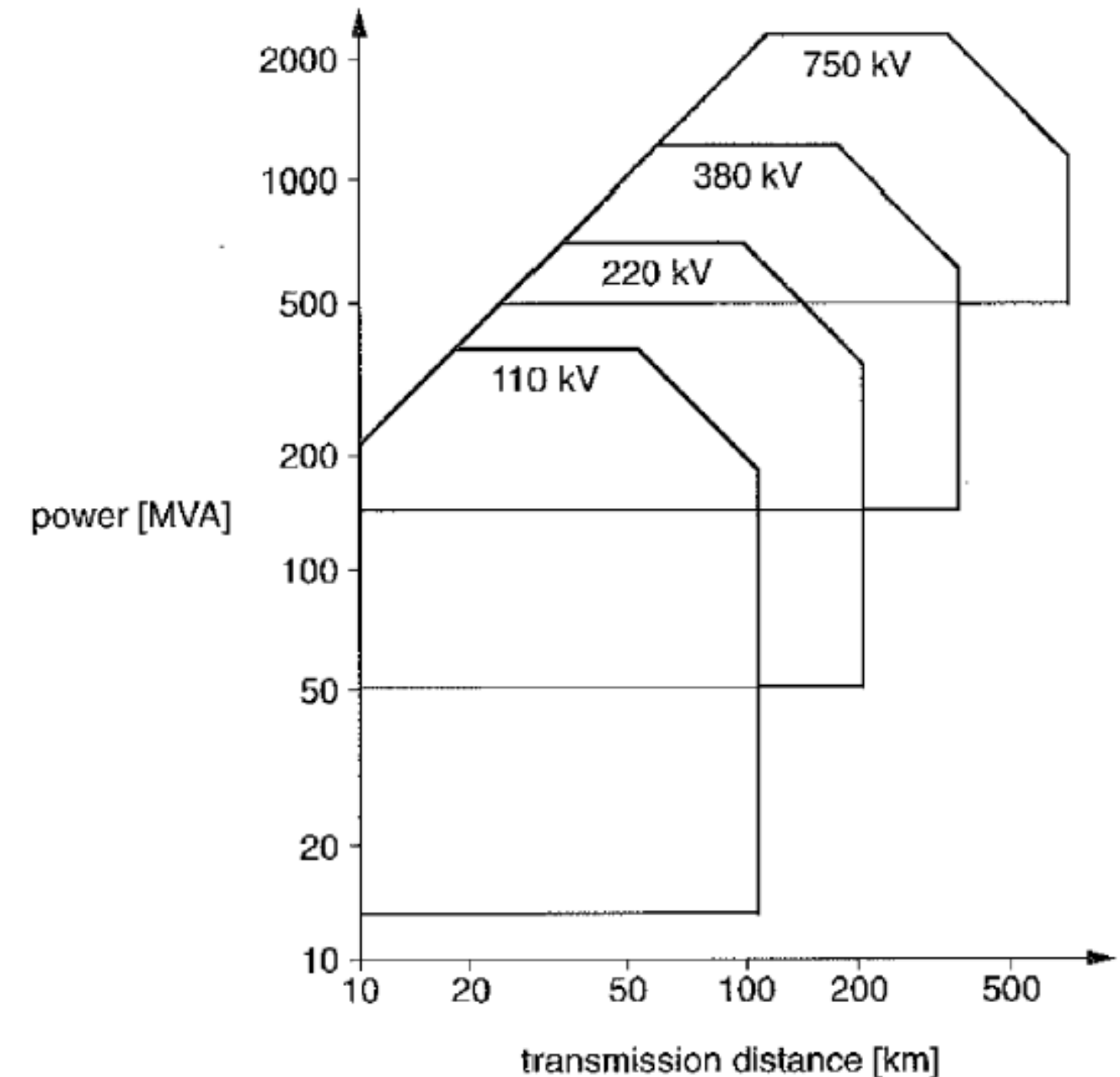
<https://www.iea.org/data-and-statistics/data-tools/real-time-electricity-tracker>

Electrical Power Transmission – Advantages

- ◆ Better system efficiency
 - total installed power can be less than sum of loads
- ◆ Improves system reliability
 - Backup generating power available
- ◆ Smaller frequency deviations
 - Great number of synchronous generators run in parallel (rotating mass)
- ◆ Cover large geographical area
 - Different peak loads because of time zones
 - Different weather conditions (Wind, Sun, ...)
- ◆ Possibility to create an electrical energy market

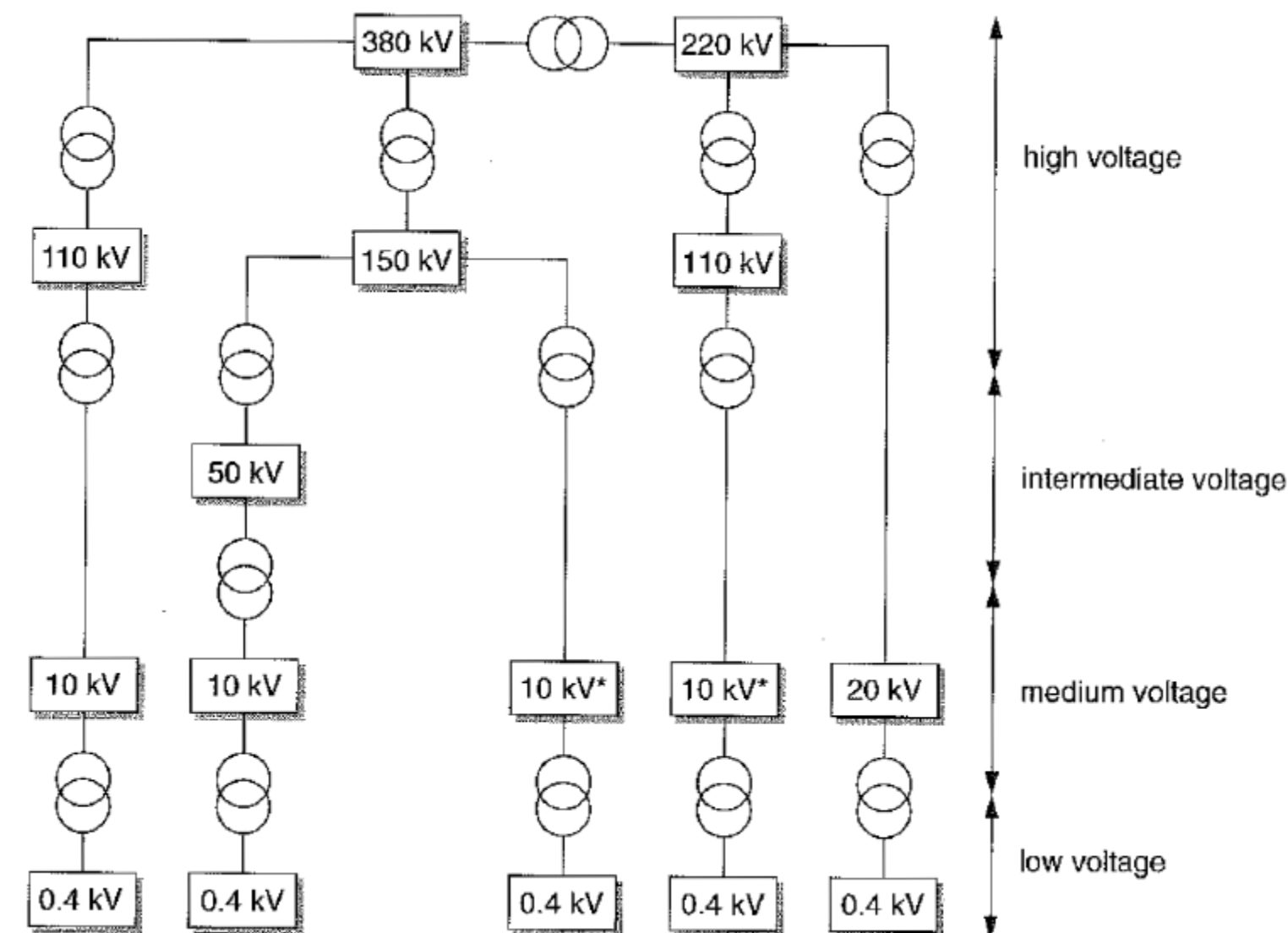
Electrical Power Transmission – Network Classification

- ◆ Separated
 - Transmission network
 - Distribution network
- ◆ Transmission network
 - High amounts of electrical energy over long distances
- ◆ Distribution network
 - Distributes within a certain area to smaller power consumers



Electrical Power Transmission – Network Structure

- ◆ Transmission network
 - Long distances
 - High amount of power
 - High voltages
- ◆ Transmission line substations
 - Close to the area of high energy consumers
 - Power transformer used
 - Feed the power into the local distribution subsystem



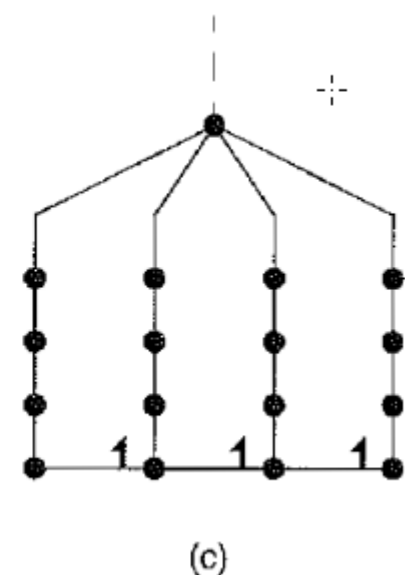
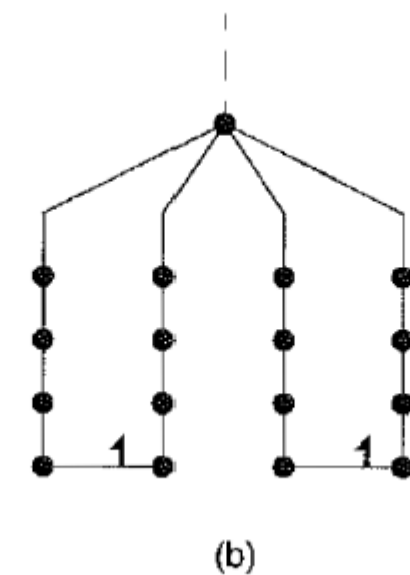
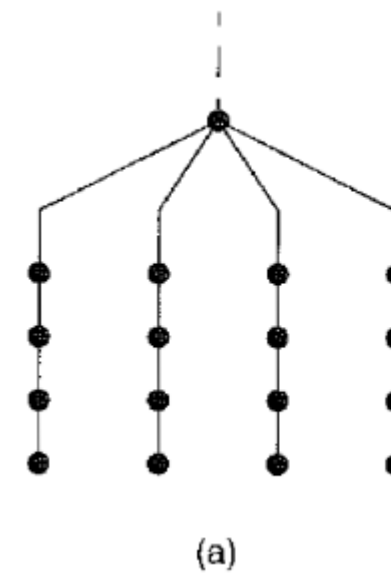
Electrical Power Transmission – Network Structure

- ◆ Transmission network consists of
 - Overhead lines
 - Underground cables
 - Power transformers

- ◆ Network can be structured as
 - Single-Point Feeding
 - Multi-Point Feeding

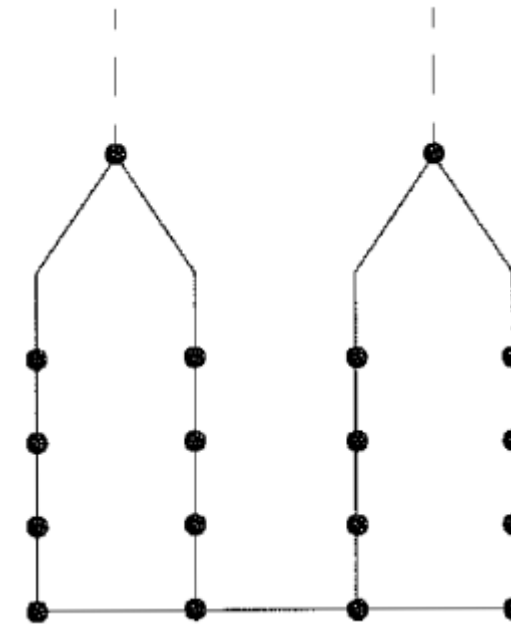
Electrical Power Transmission – Network Structure

- ◆ Single point feeding
 - Radial Structure (a)
 - One central supply
 - Directly to substations
 - Loop Structure (b)
 - Each substation fed from two sides
 - Multi Loop Structure (c)
 - Each substation is fed by more than two connections



Electrical Power Transmission – Network Structure

- ◆ Multi-point feeding
 - Usually also multi loop structure
 - Highest reliability
 - Faulty feeder gets isolated
 - Second feeder takes over
 - Typical for transmission lines



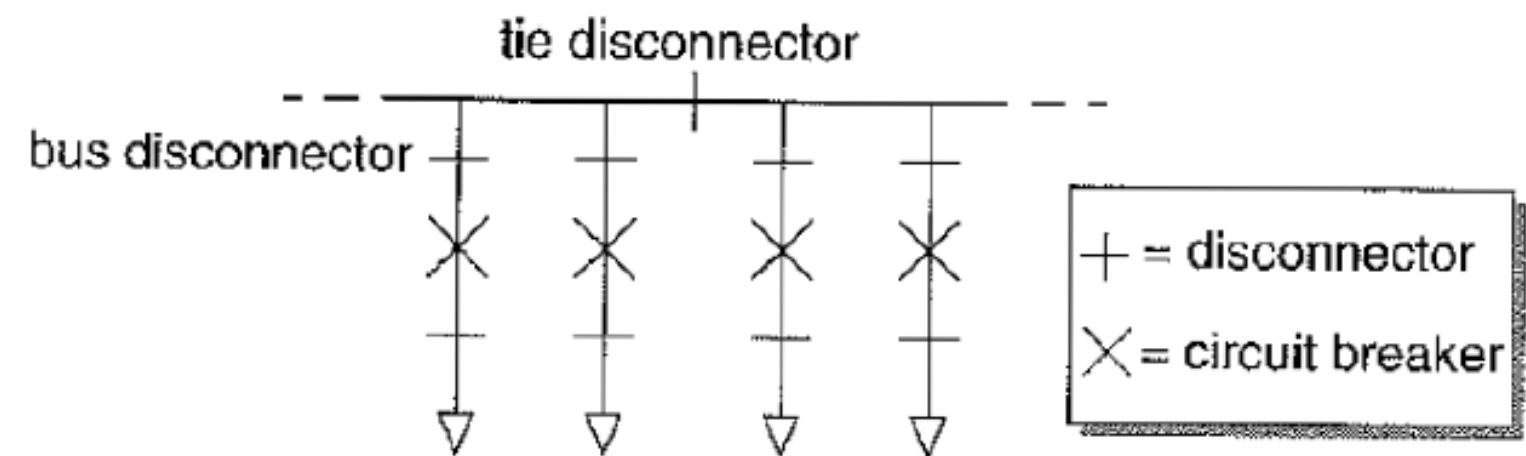
Electrical Power Transmission – Substations

- ◆ Substations
 - Nodes in the network structure
 - Interconnect different voltage levels
 - Protection equipment installed

- ◆ Substation layouts
 - Single bus system
 - Double bus system
 - Polygon bus system
 - One-and-a-half circuit breaker concept

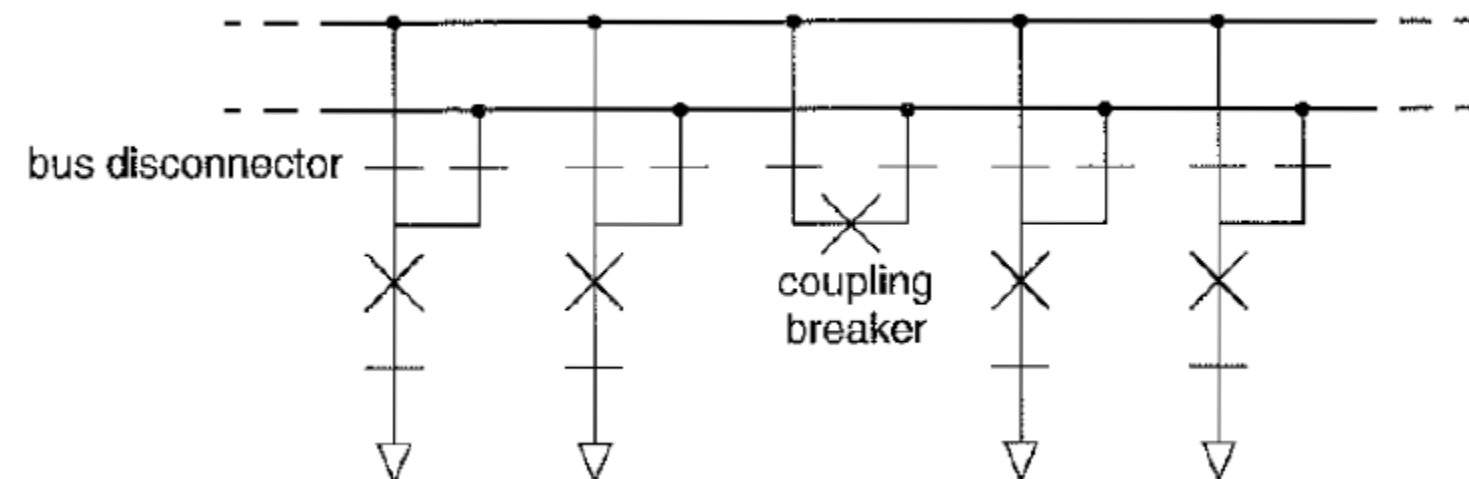
Electrical Power Transmission – Substations – Single Bus

- ◆ Bus bar supplied by incoming power line
- ◆ Feeder are connected to single bus bar
- ◆ Divided into two parts
 - Increase reliability
- ◆ Advantage
 - Easy to operate
 - Simple to protect
- ◆ Disadvantages
 - Maintenance at tie breaker
 - Entire substation or at least two feeder out of service



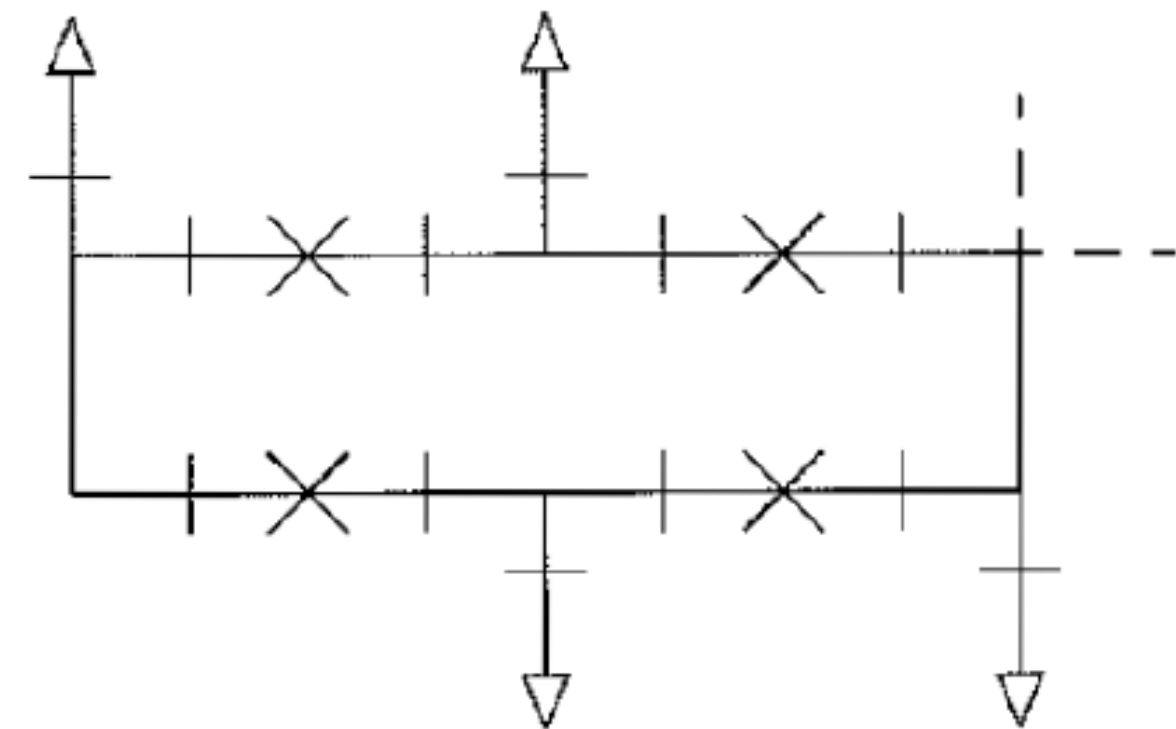
Electrical Power Transmission – Substations – Double Bus

- ◆ Most common concept
- ◆ Each feeder has two bus connections
- ◆ Busses can be coupled by coupling breaker
- ◆ Properties
 - Bus fault – feeder can be switched to healthy bus bar
 - Maintenance on bus bar without taking any feeder out of service
 - Selective protection of each feeder and bus bars
 - Only faulty coupling circuit breaker causes a disconnection of both bus bars
 - Easy to operate



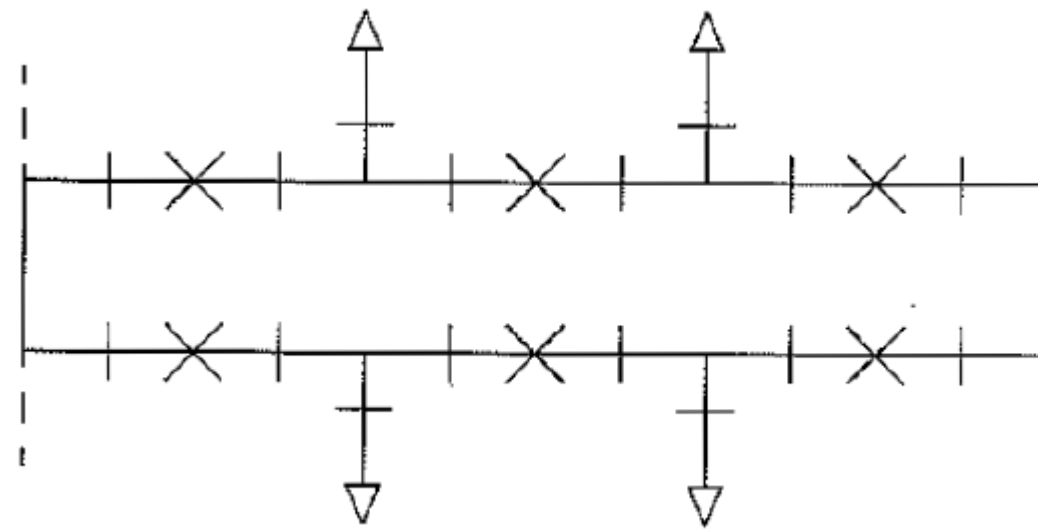
Electrical Power Transmission – Substations – Polygon Bus

- ◆ Substation concept forms a loop
- ◆ Circuit breakers are part of bus bar
- ◆ Each feeder is twice connected to the rest of the system
- ◆ Properties
 - Maintenance without disconnecting a feeder section
 - Bus fault results in disconnecting only one feeder
 - A faulty circuit breaker puts only two feeders out of service



Electrical Power Transmission – Substations – One-and-a-half circuit breaker concept

- ◆ Each feeder has an extra, shared (half) circuit breaker available
- ◆ Coupling circuit breaker missing
- ◆ Same advantages as polygon bus system



Electrical Power Transmission – Protection

- ◆ Located in substations
- ◆ Detect and clear faults
- ◆ Consists of
 - Protection relay
 - Current and voltage transformers
 - Circuit breaker

Electrical Power Transmission – Protection

- ◆ Continuously monitors the grid
 - Via current and voltage transformers

- ◆ Abnormality detected
 - Operates circuit breaker
 - Isolate faulted sections

- ◆ Abnormal situations
 - Over- Undervoltage
 - Overcurrent
 - Over- Underfrequency
 - ...

Electrical Power Transmission – Power Transformers

- ♦ Essential for transmission lines (different voltage levels)
- ♦ Generators
 - Voltage level limited to 10-25 kV because of insulation
- ♦ Transmission lines
 - In Austria up to 380 kV to transport electrical power over long distances
- ♦ Distribution lines
 - Between 10 to 30 kV to distributed between an area with many customers
- ♦ Household
 - 400 V and below

Electrical Power Transmission – Power Transformers

- ◆ The ideal transformer
 - Neglect resistance
 - Neglect leakage flux
 - Assume permeability is infinite
- ◆ Relation between Primary and Secondary Voltage:

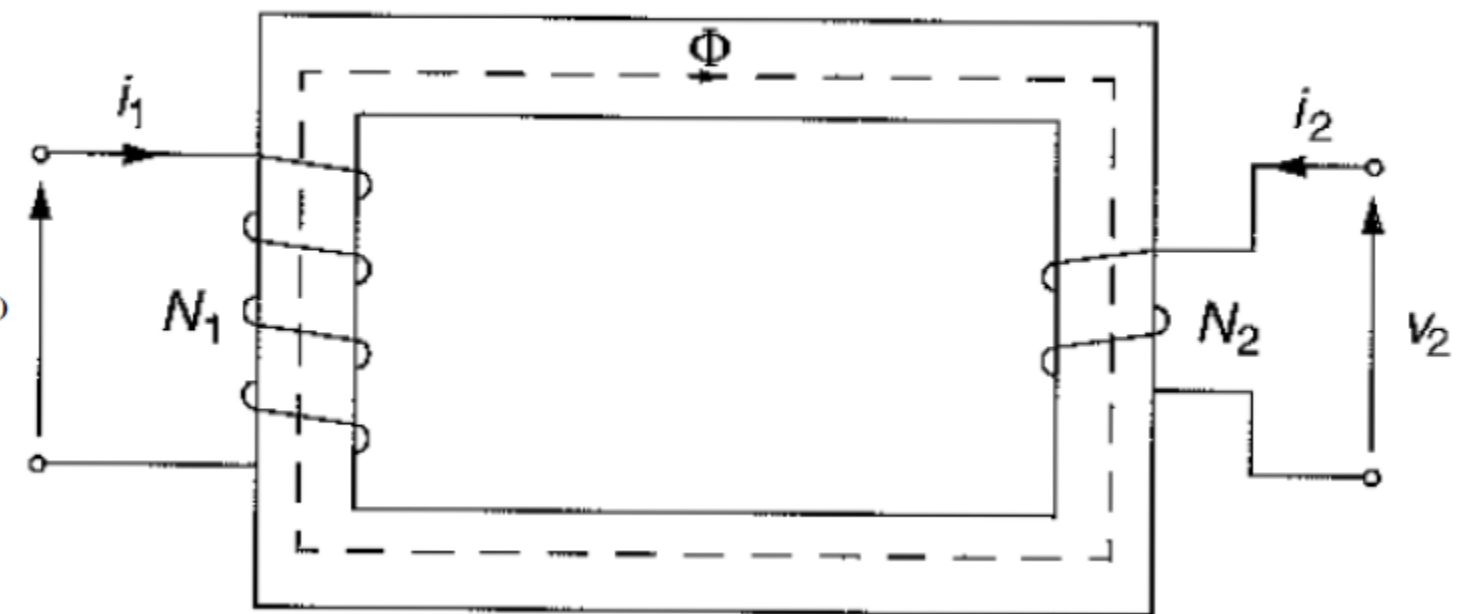
$$v_1(t) = \sqrt{2} |V_1| \cos(\omega t) = N_1 \frac{d\Phi}{dt}$$

$$\Phi(t) = C + \frac{\sqrt{2} |V_1|}{\omega N_1} \sin(\omega t) = \sqrt{2} |\Phi| \sin(\omega t)$$

$$|\Phi| = |B| A$$

$$v_2(t) = N_2 \frac{d\Phi}{dt} = \frac{N_2}{N_1} \cdot \sqrt{2} |V_1| \cos(\omega t) = \frac{N_2}{N_1} \cdot V_1(t)$$

$$\boxed{\frac{v_1}{v_2} = \frac{V_1}{V_2} = \frac{N_1}{N_2} = n}$$



Abbreviations

N_1 — Number of turns, primary transformer winding

N_2 — Number of turns, secondary transformer winding

Φ — Magnetic flux [Wb = Vs]

C — Integration constant [Wb]; zero in steady state conditions

B — Magnetic flux density [T = Wb/m²]

A — Cross-sectional area of the iron transformer core [m²]

n — turns ratio

Electrical Power Transmission – Power Carriers

- ◆ Overhead lines
- ◆ Underground cables
 - More reliable
 - Not affected by wind, lightning or car accidents
 - 6 to 10 times more expensive



www.fhv.at

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